

Varun Ramesh Jois

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Summary

PhD candidate in Computer Science with a focus on computer vision and generative AI for faces. My research focuses on how reference images can be used to improve face-related tasks from super-resolution and video enhancement to facial expression editing. Four peer-reviewed publications. Experienced in building and training deep learning models (CNNs, GANs, Diffusion Models, Transformers) using PyTorch with an emphasis on practical and deployable systems. Expected graduation August 2026.

Education

Brandeis University, Waltham, MA

Expected August 2026

PhD in Computer Science

Brandeis University, Waltham, MA

May 2020

MS in Computer Science

St. Joseph's College, Bangalore, India

April 2014

BS in Mathematics, Statistics and Economics

Publications & Research

FlameTracer: Easy and Intuitive Facial Expression Transfer and Control

Under Review, British Machine Vision Conference (BMVC) 2026 <https://github.com/varun-jois/expression-control>

- Facial expression generation is the task of producing an image with a user defined expression on an input facial image; for instance changing a person's expression from a frown to smile. Existing methods require expert knowledge of arcane facial representation systems (e.g. action units or valence-arousal coordinates) to control the intensity of the generated expression, making them impractical for everyday users.
- Designed FlameTracer, an MLP built on top of an IP-Adapter style Stable Diffusion Model (Arc2Face) for facial expression transfer. Through carefully crafted datasets, FlameTracer learned the trajectory of expressions in the 3DMM representation space for faces (FLAME) enabling smooth intensity interpolation. Crucially, the model is user friendly, needing only an additional image of the intended expression and a scalar value to control the intensity (e.g. a half smile).
- Outperformed GAN-based and diffusion-based baselines by more than 30 percentage points in a user study conducted on cloudfiresearch.com. In addition, outperformed the baselines on all objective metrics related to emotion and expression accuracy.

FSFVE: Few Shot Compressed Face Video Enhancement

Data Compression Conference (DCC) 2026, pp. 133–142 <https://github.com/varun-jois/FSFVE>

- Millions of people around the world suffer from poor quality videocalls due to limited bandwidth. Most neural based methods for improving video quality require GPUs to run making them impractical to run on typical consumer hardware.
- Designed an instance-specific model that trains on as few as 10-30 frames of a subject in under 100 seconds. The model avoided the speed bottleneck of CNNs by working on non-overlapping 8x8 patches. These patches were converted into the frequency domain and then passed to a sine activated neural network which allowed for fast training convergence. During training, the low-frequency coefficients were given greater weight to prioritize the structures most critical to visual quality.
- Achieved a BD-Rate of -25.12% and a BD-PSNR of $+1.33$ dB over the H.264 input video outperforming all baselines across all compression settings for both H.264 and H.265. Performed in realtime at 30.24 FPS on a typical laptop CPU.

RTFVE: Realtime Face Video Enhancement

Computer Analysis of Images and Patterns (CAIP) 2025, pp. 136–148 github.com/varun-jois/RTFVE

- Similar to the above paper, this work sought to improve the quality of videocalls by building a model that ran in realtime on typical consumer CPUs but took a different approach by leveraging high-quality reference frames from earlier in the call.
- Built a dual-branch CNN-GAN that takes a compressed frame and 1-3 high-quality reference frames as input, to produce the enhanced frame. Since the references come from the same call, they are very similar to the incoming frames and can be used as a prior for the enhancement. The reference features are cached by the model allowing for faster inference. The model was designed to be integrated with current video compression standards.

- Achieved realtime inference at 24.85 FPS on a standard laptop CPU with only 190K parameters (<1 MB). The first neural face enhancement model to meet realtime CPU requirements with PSNR improvements of up to 0.8 dB over the H.264 baseline.

Reference-Based Face Super-Resolution Using the Spatial Transformer

Asian Conference on Computer Vision (ACCV) 2024, pp. 3689–3705 github.com/varun-jois/FSRST

- Reference-based Face Super-Resolution is the task of increasing the resolution (and fine details) of a facial image with the help of high resolution reference images of the same person. Using reference images can greatly improve output quality, but aligning references to the input is challenging and existing methods based on deformable convolutions suffer from training instability.
- Introduced a new alignment module based on the spatial transformer that is considerably more stable than deformable convolution alignment. To aggregate signals from the references, we developed a new aggregation technique based on the L2-distance that weights references by similarity, gracefully falling back to single-image SR when no good references are available.
- Achieved state-of-the-art PSNR on three datasets: 32.08 dB on DeepFakeDetection, 29.28 dB on CelebAMask-HQ, and 31.75 dB on VoxCeleb2 outperforming models with up to 10× more parameters.

Skills

Machine Learning	CNNs, GANs, Diffusion Models, Transformers, RNNs, Genetic Algorithms
Computer Vision	Super-Resolution, Video Compression, Face Enhancement, Generative AI
NLP	Named Entity Recognition, Machine Translation, Information Retrieval
Programming	Python, Java, SQL
Frameworks & Tools	PyTorch, OpenCV, NumPy, scikit-learn, pandas, Matplotlib, Captum
Miscellaneous	Git, Bash, FFmpeg, Slurm, L ^A T _E X

Industry Experience

Analytics Consultant / Data Scientist · Emplay Inc, Bangalore, India April 2017 – June 2018

- Collaborated with the product team to build Emplay’s data-driven sales enablement product, Sales Bot. Wrote production-grade algorithms that harnessed external web data to find leads. Our clients doubled the number of accounts in their territory and increased lead generation by 20
- Created convolutional neural network models (CNNs) to categorize text data using the GloVe embeddings. Improved accuracy on non-deep learning models by 15
- Designed web scraping and data pipeline infrastructure using Python (requests, selenium, redis) to keep databases updated daily; automated text standardisation with NLTK, saving 50-100 hours per month.

Associate Analyst · Emplay Inc, Bangalore, India April 2015 – March 2017

- Implemented the ranking algorithm crucial to Emplay’s Deal Coaching product. This contributed substantially to Emplay’s bottom line helping us grow from a team of 8 to a team of 30 within 3 years.
- Built random forest and neural network models to predict the probability of winning a deal. Improved the conversion ratio of our clients by 7
- Wrote stored procedures on Microsoft SQL Server to automate Emplay’s Rep and Deal DNA reports. Cleaned, combined, and manipulated data to spot trends saving 40-80 hours per month.

Teaching Experience

Teaching Assistant · Brandeis University, Walham, MA

Led weekly recitation classes for Advanced Programming Techniques in Java; served as TA for Data Structures, Theory of Computation, Web Application Development, and Algorithms.